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Self-calibrating Fourier ptychographic microscopy using automatic differentiation

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Fourier ptychographic microscopy (FPM) can provide high-throughput imaging by computationally combining low-resolution images at different spatial frequencies within the Fourier domain. The core algorithm for FPM reconstruction draws upon phase retrieval techniques, including methods such as the ptychographic iterative engine (PIE), regularized PIE (rPIE), and embedded pupil function FPM (EPRY-FPM). The calibration of the physical setup plays a crucial role in the quality of the reconstructed high space-bandwidth product (SPB) image. Despite advances, many methods, incorporating either machine learning or calibration algorithm, face challenges. These include the need for extensive parameter tuning and extra optical system information, hindering their practical use. To address these limitations, we introduce a novel, to the best of our knowledge, self-calibrating FPM reconstruction approach that utilizes automatic differentiation. This method diverges from traditional iterative phase and amplitude updates, opting instead to simultaneously recover a complex 2D image and refine the optical system's physical parameters. Our approach matches the effectiveness of existing recovery techniques while significantly reducing the calibration burden. In this report, we will demonstrate our method is capable of self-calibrating without needing extra system information. We validate our algorithm's performance through numerical simulations and then show its practicality by reconstructing a full field of view of cervical cell slides using ultraviolet Fourier ptychographic microscopy (UV-FPM). © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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Introduction. For the past decade, Fourier ptychographic microscopy (FPM) has grown strong interests in both academia and industry. The clever idea of using phase retrieval with frequency shifted images to recover a 2D high-throughput complex image was first introduced in 2013 by Zheng *et al.* [1,2]. Subsequently, numerous methods have been developed to enhance the accuracy and capability of FPM by considering the pupil functions of the optical system [3,4], leveraging deep learning [5,6], and making hardware improvements [7]. However, typical FPM algorithms, such as regularized ptychographic iterative

engine (rPIE) [8] and embedded pupil function recovery algorithms [3], heavily rely on the precise matching between the input physical parameters and the actual optical system. People have tried to reduce this tedious process by using information of bright- to dark-field transition [9–11] and systematically optimizing the optical system [12–14]. They either require a more robust algorithm [15] to accurately extract additional information from the image. This has posed a barrier to applying FPM in more practical setups for biomedical imaging applications. We present an alternative method to overcome this problem. With our method, we will show significant improvements in mitigating this issue, thereby enhancing the potential overall usability of FPM in practical biomedical imaging [16,17]. In recent years, automatic differentiation (AD) [18], an emerging method for automatically computing derivatives, has benefited the majority of deep learning applications [19] where it is necessary to calculate the gradient of the cost function for hundreds of mathematical operations. AD addresses this tedious issue by automatically calculating analytic partial derivatives of any function with respect to the independent variables via computational graph. This approach not only prevents numerical errors from numerical differentiation methods but also reduces the burden to derive the analytical solution of gradients. People have tried to use AD for computational imaging to improve the performance [20] and as a replacement for PIE [21,22]. In the following sections, we will introduce a novel self-calibrating FPM technique that employs AD to effectively overcome such calibration challenges. This algorithm should be also applicable to any traditional FPM experimental setup.

Methods. FPM forward model. The forward model of FPM has been comprehensively described in the work of Zheng *et al.* in 2016 [23]. In this work, we have constructed our forward model and simulations based on the physical principles of the FPM, shown in Fig. 1. The object we are trying to recover can be seen as a 2D complex matrix. The imaging process of illuminating the sample at different angles can be described by the following equation:

$$I_{LRi} = |\mathcal{F}^{-1} [O(k_x - k_{xi}, k_y - k_{yi}) \cdot CTF(k_x, k_y)]|^2, \quad (1)$$

where I_{LRi} give as the raw intensity measurements of low-resolution image and $O(k_x, k_y)$ is the Fourier transform of the object $o(x, y)$. $CTF(k_x, k_y)$, cohere transfer function can be

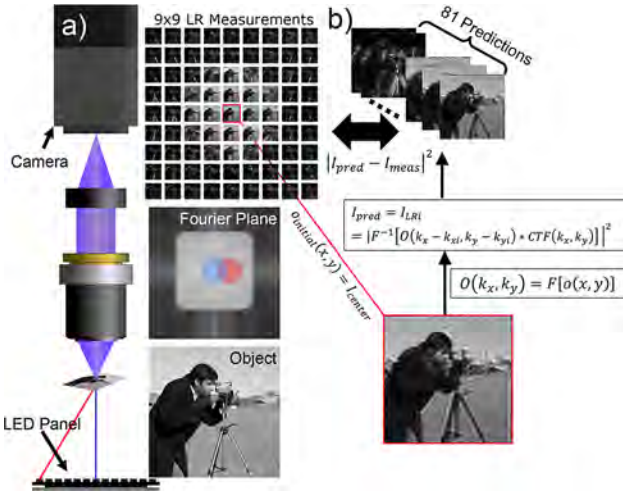


Fig. 1. FPM forward model and AD reconstruction. (a) In the FPM setup, an LED illuminates the sample from different angles, producing a series of low-resolution images with shifted frequencies. (b) AD optimization. Starting with an up-sampled center LED image as initial guess, AD minimizes the absolute error between predicted and simulated low-resolution images.

expressed as follows:

$$CTF(k_x, k_y) = LP(NA \cdot k_0) e^{i\phi(k_x, k_y)}. \quad (2)$$

$LP(\cdot)$ is the 2D-lowpass filter operator that the cutoff frequency is determined by the objective NA and the wavenumber k_0 . ϕ , the pupil function of the optical system, is characterized by Zernike modes. To convert x , y , and z into k_x and k_y , we use the following equations:

$$k_x = -k_0 \sin \left[\tan^{-1} \left(\frac{x}{z} \right) \right], k_y = -k_0 \sin \left[\tan^{-1} \left(\frac{y}{z} \right) \right]. \quad (3)$$

The rPIE algorithm recovers the object and pupil function of the optical system by updating each other alternately. However, the entire reconstruction process highly relies on the calibration of each frequency shifts k_{x_i} and k_{y_i} in the Fourier plane. This posed a challenge to many practical applications.

Self-calibrating FPM. We utilize AD to recover the image while refining the physical hyperparameters that have critical impact to the final results. The acyclic computational graph

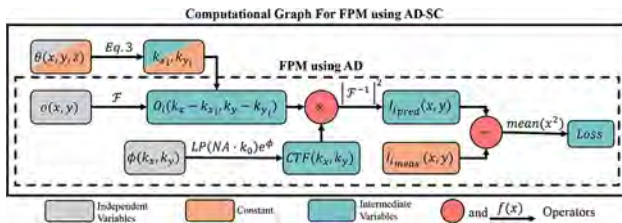


Fig. 2. AD and AD self-calibration (AD-SC) traversal for FPM reconstruction. $o(x, y)$ and $\phi(k_x, k_y)$ are the unknown object and pupil function. The FPM forward model can be decomposed into intermediate variables (shown in blue) and operators (shown in red) with known analytical derivatives. Gradient descent optimization uses the chain rule to backpropagate partial derivatives. In the direct AD approach, the LED positions (θ) are treated as constants, making k_{x_i} and k_{y_i} fixed. In the AD-SC approach, θ is treated as a variable, so k_{x_i} and k_{y_i} become intermediate variables.

of our forward model is described in Fig. 2. There are various gradient descent methods developed over time such as stochastic gradient descent (SGD), AdaGrad, and RMSProp. We choose Adam [24], which is a robust method that combines the advantages of AdaGrad and RMSProp by adaptively estimating the lower-order moments of gradients. The merit function for recovering the high-resolution image and refining the physical parameters can be written as follows:

$$\mathcal{L} = |f(I_{HR}, \theta) - I_{LR}|^2. \quad (4)$$

θ are the hyperparameters that contain the x , y , and z shifts of the LED positions. $f(\cdot)$ is the operator of the FPM forward model. For direct AD reconstruction, as depicted in Fig. 2, we do not optimize θ . We update only the object and pupil function at every iteration while being constrained by Eq. (4). For AD self-calibration (AD-SC), we add θ into the optimization process. θ represents the individual x - y - z positions of each LED. In this way, we update LED positions iteratively, which will reflect as corrections of frequency shifts (k_{x_i} and k_{y_i}) of each raw measurement in the Fourier domain. In the Fourier domain, the overlapping region between adjacent measurements must be

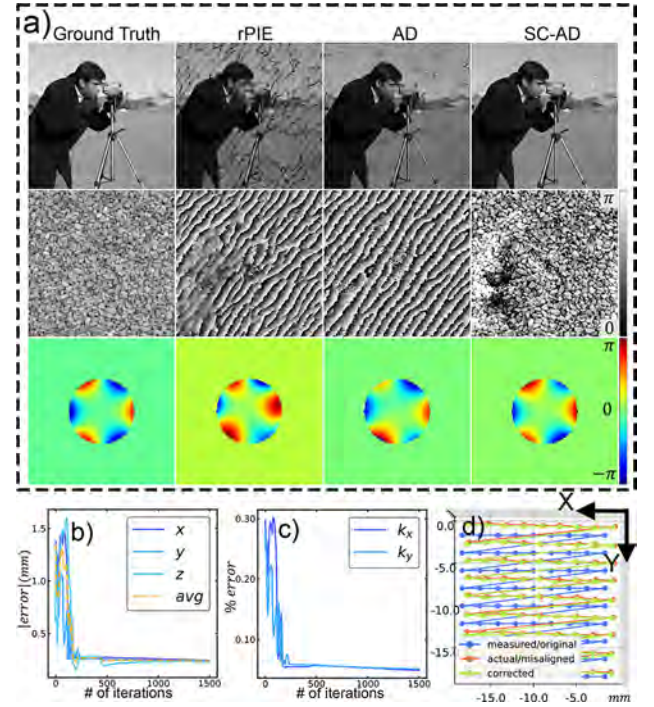


Fig. 3. Simulation results. (a) From left to right, images show the ground truth, followed by those recovered by rPIE, AD, and AD self-calibration (AD-SC). From top to bottom, they show the amplitude, phase, and pupil function used for the simulation. (b) Absolute error in x , y , and z LED positions (in mm) versus iterations. (c) Percentage error in k_x and k_y versus iterations. (b) and (c) demonstrate that AD-SC iteratively recovers the positions of the LEDs and, more importantly, their corresponding shifts in the Fourier domain, k_x and k_y , respectively. The oscillations at the start of the iterative process is due to ambiguity presented in Eq. (3). k_x and k_y are determined by the ratio of x/z and y/z rather than the absolute spatial positions of the individual LEDs. Position errors should eventually converge due to the overlapping information between the captured images. (d) Top view of LED positions, with red for misaligned positions, blue for initial inputs, and green for corrected positions, connected by a simulated zigzag scanning route.

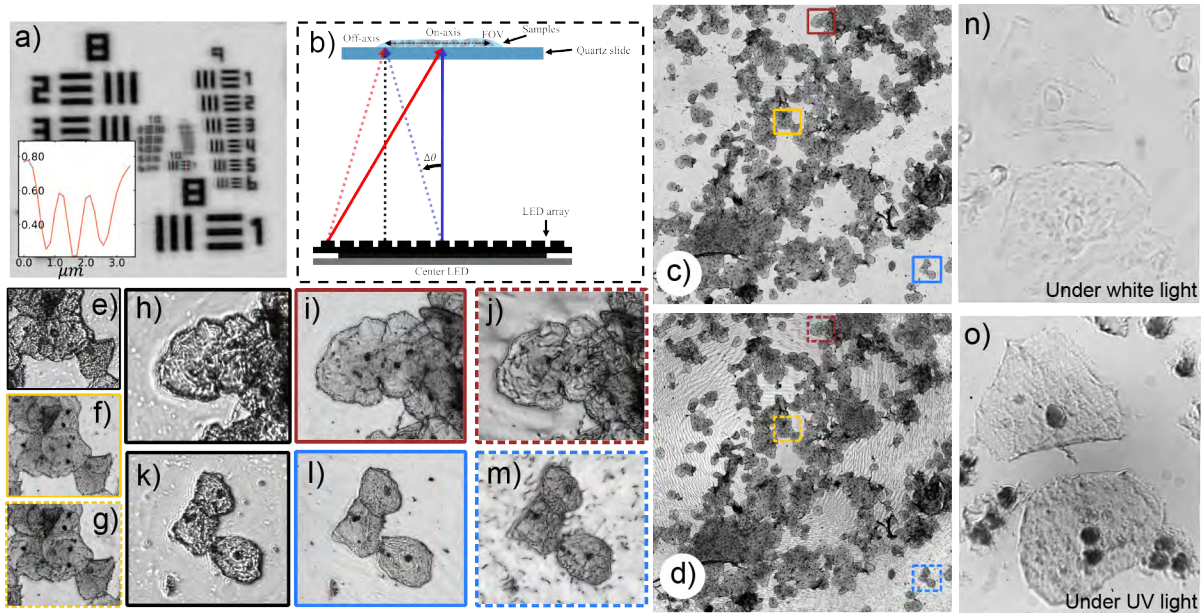


Fig. 4. UV-FPM experimental results. (a) Recovered USAF target shows UV-FPM is capable of achieving a resolution limit up to 486 nm (first element of the tenth group). (b) Illumination angle differs between off-axis and on-axis regions for the same LEDs. This means LED positions need to be re-calibrated to recover high-resolution images of off-axis areas. (c) AD-SC recovered results that self-calibrate the LED position at off-axis regions where we do not see the artifacts caused by the miscalibration compared to (d). (d) Traditional FPM recovery without any corrections of LED positions at off-axis regions. We see significantly degraded recovered results in those regions such as enlarged in (j) and (m). (i) and (l) The same regions recovered by AD-SC showed significant improvements. (h) and (k) Raw images at the same off-axis regions. (f) and (g) On-axis recovered images by AD-SC and the traditional recovery method, respectively, as a reference. (e) Captured low-resolution raw image at the same area. (n) and (o) show the same cervical cells under white light illumination with 40×0.65 NA and under UV light with 50×0.42 NA.

satisfied. This means the relative frequency shifts between each adjacent measurements is well-constrained. Therefore, AD-SC can accurately recover the object and shifted LED positions at the same time. We tried adding regularization terms such as L1 norm and total variation (TV) to Eq. (4), but they did not significantly improve the recovered high-resolution image. These terms also introduce extra tuning parameters, making the reconstruction process more complex, as different samples may require different weights for regularization due to their unique characteristics. We implemented the described AD and AD-SC FPM reconstruction methods using PyTorch.

Simulation. We verified AD-SC by numerically introducing misalignment into the system. Then, we compared the results of our self-calibrating reconstruction with those obtained using the current rPIE method. The main goal of this misalignment simulation was to test the ability of AD-SC to correct LED positions by introducing uncertainties that may arise during the experimental procedure. These uncertainties include misalignment of the center LED in the LED array and misplacement of LED positions during manufacturing. We introduced x - y - z shifts and rotations to the entire LED panel and individual random x - y - z shifts to every LED. With the misaligned LED positions, we generated the corresponding low-resolution raw images. Then, we used the original LED positions, as one would for an actual FPM experiment, as input to the algorithm for reconstruction. We compared recovered images and pupil functions in Fig. 3(a). The rPIE method iteratively updates the 2D complex object and pupil function simultaneously, where the pupil is also constrained by Zernike modes but does not update the input physical parameters at all. If the system is miscalibrated by a small amount, the

recovered image will be significantly impacted. In the direct AD approach, we only optimize our recovered 2D complex image and pupil function without optimizing the LED positions. In this way, we recovered the amplitude of the image but not the phase. We think this is because the loss function is established based on the intensity measurements of the system only (Eq. (1)), without constraining the phase of the object. AD-SC optimized the LED position in each iteration to obtain a much more accurate result compared to the previous two methods. This error reduction became more obvious when demonstrated in our experiments. We showed that AD-SC significantly outperforms the traditional FPM algorithm in Fig. 3. AD-SC can correct the LED spatial positions to the actual/misaligned positions with small variations (Figs. 3(b) and 3(d)). This is because FPM combines images in the Fourier domain instead of the spatial domain, making it more accurate to monitor errors with k_x and k_y (Fig. 3(c)).

Experiment and results. We used a self-assembled UV LED board with a peak wavelength of 265 nm and an 10 nm bandwidth, mounted on an x - y motorized stage (two PI Q-545.240) for illumination. We employed a 5×0.12 NA UV objective lens and captured images with a 2840×2840 , $2.74\ \mu\text{m}\times 2.74\ \mu\text{m}$ pixel size, Lucid Vision UV camera (Atlas10 UV 8.1 MP). The sample was positioned 95 mm from the UV LED board, and 121 images were taken with a 3.5 mm LED spacing. The theoretical resolution of our UV-FPM [25] is about 431 nm with a synthetic NA of 0.307. The measurement with a USAF target was about 486 nm shown in Fig. 4(a), which matches our expectations. We demonstrate the difference between AD-SC and conventional FPM by recovering the entire FoV of an unstained cervical slide, as shown in Figs. 4(c) and 4(d). Unstained cervical cells

are directly collected from patients and smeared onto a quartz slide for imaging. With the unique absorbance of cellular nucleic acids at 260 nm, we could clearly see the nuclei under UV illumination. A significant challenge associated with practical experiments is the off-axis region of a large FoV, where the LED positions relative to the off-axis regions no longer align with those in the on-axis region, as shown in Fig. 4(b). That is, the initial LED positions used for recovering the on-axis region of the image become incorrect when trying to recover the off-axis region. This situation is analogous to misaligned LED positions. The misalignment at the off-axis region increases as the field of view (FoV) expands. For example, in a $2\text{ mm} \times 2\text{ mm}$ FoV, the center LED would be misaligned by 1 mm at the edge of the FoV. We can observe that using the initial LED positions to reconstruct off-axis regions results in strong reconstruction artifacts, as shown in Fig. 4(d). We also enlarged off-axis regions, as illustrated in Figs. 4(j) and 4(m). Hence, to recover high-resolution image in off-axis regions compared to on-axis areas, the illumination angles must be adjusted by $\Delta\theta$ degrees, which varies depending on the distance of the off-axis region from the center. This misalignment inevitably hinders the practical usage of FPM. A common workaround has been to assume that the center of the image is perfectly aligned with the center LED on the LED panel, which is often not the case in reality. Then, we can use relative pixel distances between the off-axis region and the center pixel on the captured image to re-calibrate LED positions. However, this solution still heavily relies on the calibration of the optical system by assuming that all optical elements and the center of the sensor are perfectly on-axis. With AD-SC, we can recover a full FoV image without needing to re-calibrate LED positions for each off-axis region across the entire FoV. Compared to on-axis recovered results, as depicted in Figs. 4(f) and 4(g), both AD-SC and rPIE showed the capability to recover high-resolution image accurately which matches with Our expectation. For off-axis regions, AD-SC effectively eliminates the reconstruction artifacts caused by miscalibration. This improvement is crucial for practical implementation of FPM in biomedical imaging.

Discussion and conclusion. In this work, we presented a self-calibrating FPM approach that paves the way for practical biomedical imaging applications. We demonstrated the use of AD-SC for reconstructing high-throughput, label-free cervical cell images with the ability to self-calibrate LED positions. The primary advantage of using FPM is its ability to maintain a large FoV while enhancing resolution. With our method, re-calibrating the LED positions to recover the full FoV is unnecessary, allowing accurate image recovery even if the LED panel is not perfectly aligned. This optimization process is reliable, and the flexibility within the AD optimization process could be further improved to reduce computational resources. We believe

our methods demonstrate the increased potential for FPM to be used in more practical applications.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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